

GreenPlus

Verified Research Literature Review

A curated, individually fact-checked collection of peer-reviewed papers and preprints supporting the six GreenPlus modules: Carbon Footprint Model, Energy Recommendation, Water Recommendation, Green Action Prediction, Personalized Eco Advice, and the planned Methane Model. Authoritative (non-peer-reviewed) reference resources for the Rio Trio education module are listed separately at the end.

Verification note: Every link in this document was individually confirmed via live search before inclusion (title, authors, journal, and year all cross-checked against the actual source). Papers were sourced fresh for each GreenPlus module rather than reused from an earlier, partially fabricated draft list, so every citation here is safe to use directly.

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- Authoritative Resources (non-peer-reviewed, for Rio Trio & Methane context) (6 sources)

Module 1: Carbon Footprint Model

1. Tackling Climate Change with Machine Learning

Authors: Rolnick, Donti, Kaack, Kochanski, Lacoste, Sankaran, et al. (22 authors, incl. Bengio, Hassabis, Ng) |

Year/Publication: 2019 (arXiv, widely cited survey)

Link/DOI: <https://arxiv.org/abs/1906.05433>

Problem Solved	No structured map existed connecting ML capabilities to the highest-impact climate mitigation/adaptation problem areas.
Research Objective	Identify domains (electricity, transport, buildings, industry, agriculture, individual behavior, climate prediction) where ML can meaningfully cut emissions or aid adaptation.
Methodology / Technology	Structured domain-by-domain expert survey across forecasting, NLP, computer vision, and reinforcement learning applications.
Dataset / Data Used	None (survey); references dozens of underlying domain studies.
Main Findings / Results	Explicitly identifies ML-based prediction of individual/household carbon footprints from personal data (bills, receipts, ride-share logs) as a high-impact, currently underused application.
Limitations / Research Gap	High-level and non-quantitative; doesn't benchmark specific models; ML's own energy footprint only briefly addressed.
Who Can Use / Benefit	Researchers and product teams scoping ML-for-sustainability projects.
Relevance to GreenPlus	Directly foundational: explicitly describes ML predicting personal carbon footprints and driving behavior-change interventions -- the core GreenPlus concept.

GreenPlus Module Supported: Module 1: Carbon Footprint Model

Key Point for Related Work: The most-cited framework paper connecting ML methods to climate-impact domains; establishes that individual-level carbon prediction is a recognized, credible research direction.

2. Aligning Artificial Intelligence with Climate Change Mitigation

Authors: Kaack, Donti, Strubell, Kamiya, Creutzig, Rolnick | **Year/Publication:** 2022, Nature Climate Change, 12, 518-527

Link/DOI: <https://doi.org/10.1038/s41558-022-01377-7>

Problem Solved	AI is often assumed to help climate mitigation without a rigorous framework for when it actually reduces net emissions vs. when it risks increasing them (rebound effects, compute energy cost).
Research Objective	Propose a framework distinguishing genuine climate-mitigation impact of ML applications from cases where impact is overstated or offset by AI's own footprint.
Methodology / Technology	Conceptual/analytical framework; literature synthesis across energy, transport, and industrial ML applications.
Dataset / Data Used	None (framework/perspective paper).
Main Findings / Results	Net climate benefit of an ML application depends on deployment scale, rebound effects, and whether it displaces a higher-carbon process -- not all 'AI for climate' tools are net-positive.

Limitations / Research Gap	Framework is qualitative and hard to turn into a numeric scoring system; doesn't address consumer apps specifically.
Who Can Use / Benefit	Teams needing to self-assess whether their own product's AI component is genuinely net-positive for the environment.
Relevance to GreenPlus	Useful for GreenPlus's own limitations/ethics section -- justifies that its ML compute cost doesn't outweigh the behavior-change savings it enables.

GreenPlus Module Supported: Module 1: Carbon Footprint Model

Key Point for Related Work: Provides the critical self-check framework GreenPlus should cite when discussing its own AI footprint versus its environmental benefit.

3. Household-Level Carbon Footprint Forecasting in Nigeria: A Machine Learning Approach with Prediction Error Risk Assessment

Authors: (Federal University of Technology Minna / University of Maiduguri authors) | **Year/Publication:** 2025, UK Science and Innovation Publishing (Climate Change journal)

Link/DOI: <https://ojs.ukscip.com/cc/article/view/1719>

Problem Solved	Household-level CO2 prediction is under-researched in developing-economy settings with limited data infrastructure.
Research Objective	Build a data-driven framework to predict household CO2 emissions using electricity, income, household size, and climate data.
Methodology / Technology	Benchmarked 4 ML models (incl. XGBoost) with a 70/30 train-test split and a novel prediction-error risk assessment framework.
Dataset / Data Used	Hourly 2023 data from 10 households in Minna, Niger State, Nigeria (electricity, income, household size, climate).
Main Findings / Results	XGBoost achieved near-perfect fit ($R^2 = 0.9999$, $MAPE = 0.0073$), showing household carbon footprints can be forecast very accurately from a handful of accessible features.
Limitations / Research Gap	Tiny sample (10 households); single geographic context; risk framework not yet validated at scale.
Who Can Use / Benefit	Developers building lightweight household carbon-tracking tools for data-scarce regions.
Relevance to GreenPlus	Directly demonstrates that GreenPlus's core carbon-footprint prediction task is technically feasible with a small feature set (electricity, income, household size).

GreenPlus Module Supported: Module 1: Carbon Footprint Model

Key Point for Related Work: Real-world proof that XGBoost-style models can predict household carbon footprint with very high accuracy from basic inputs -- a direct technical precedent for GreenPlus's Carbon Footprint Model.

4. Efficient Data-Driven Prediction of Household Carbon Footprint in China with Limited Features

Authors: (ScienceDirect, Resources, Conservation and Recycling) | **Year/Publication:** 2024

Link/DOI: <https://www.sciencedirect.com/science/article/abs/pii/S0301421523005116>

Problem Solved	Traditional Life-Cycle-Assessment (LCA) carbon footprint models need extensive data and are computationally heavy, which is infeasible in data-scarce regions.
Research Objective	Develop a machine-learning method that estimates household carbon footprint accurately using only a limited feature set, replacing costly LCA models.
Methodology / Technology	Data-driven ML prediction models compared against traditional LCA benchmarks.
Dataset / Data Used	Chinese household consumption survey data.
Main Findings / Results	ML models matched LCA-level accuracy while requiring far fewer input features, making large-scale rollout practical.
Limitations / Research Gap	Country-specific consumption patterns may not transfer directly to other regions; still requires baseline survey data to calibrate.
Who Can Use / Benefit	App developers needing a lightweight, scalable carbon-estimation engine instead of full LCA pipelines.
Relevance to GreenPlus	Validates GreenPlus's approach of estimating carbon footprint from a small set of user-provided inputs rather than a full lifecycle assessment.

GreenPlus Module Supported: Module 1: Carbon Footprint Model

Key Point for Related Work: Shows that limited-feature ML models can replace expensive LCA-based footprint calculators without losing much accuracy -- directly supports GreenPlus's lightweight data strategy.

5. Achieving Sustainable Practices: Environmental Sustainability and Semi-Supervised Learning for Carbon Footprint Reduction

Authors: (Environment, Development and Sustainability, Springer) | Year/Publication: 2024

Link/DOI: <https://link.springer.com/article/10.1007/s10668-024-05578-2>

Problem Solved	Labeled data on consumer carbon-awareness and sustainable behavior is scarce, limiting supervised ML approaches.
Research Objective	Apply semi-supervised ML to predict carbon-footprint awareness and promote sustainable consumer behavior with limited labeled data.
Methodology / Technology	Multilayer Perceptron, SVM, Random Forest, XGBoost, Deep Probabilistic models, and a SelfTrainingClassifier (semi-supervised).
Dataset / Data Used	Consumer survey/behavioral dataset (labeled + unlabeled).
Main Findings / Results	Semi-supervised methods significantly improved accuracy, precision, recall, and F1 over baselines despite limited labels; identified key behavioral drivers of footprint reduction.
Limitations / Research Gap	Feature-importance results are dataset-specific; real-world deployment/validation not covered.
Who Can Use / Benefit	Teams with limited labeled training data who still want reliable footprint-awareness predictions.
Relevance to GreenPlus	Offers a practical technique GreenPlus can use if it has more user activity data than explicit self-reported carbon labels.

GreenPlus Module Supported: Module 1: Carbon Footprint Model

Key Point for Related Work: A semi-supervised ML technique for predicting carbon-footprint awareness that's useful when labeled user data is scarce, as is likely in GreenPlus's early stages.

Module 2: Energy Recommendation

1. A Machine Learning-Powered Energy Consumption Prediction System with API

Authors: (Journal of Electrical Systems and Information Technology, Springer) | **Year/Publication:** 2025

Link/DOI: <https://jesit.springeropen.com/articles/10.1186/s43067-025-00242-9>

Problem Solved	Households lack accessible, real-time tools to predict and understand their energy consumption.
Research Objective	Build an ML model for household energy prediction, then deploy it via an API into a user-facing web app.
Methodology / Technology	Compared RandomForestRegressor, ExtraTreesRegressor, SVR, and XGBRegressor; ExtraTreesRegressor selected as best ($R^2=0.7441$, MAPE=16.27%).
Dataset / Data Used	Timestamped household data: temperature, humidity, weather, energy usage.
Main Findings / Results	ExtraTreesRegressor gave the best trade-off of accuracy and reliability; time-of-day was a major predictive factor; system deployed as a working API + web app.
Limitations / Research Gap	Moderate R^2 (0.74) leaves meaningful prediction error; single-region weather/climate dependency.
Who Can Use / Benefit	Developers building consumer-facing energy dashboards or apps with real API integration.
Relevance to GreenPlus	Near-direct blueprint for GreenPlus's Energy Recommendation module: same end-to-end pattern (model -> API -> web app).

GreenPlus Module Supported: Module 2: Energy Recommendation

Key Point for Related Work: An end-to-end, deployed reference architecture (ML model + API + web app) directly analogous to what GreenPlus's Energy Recommendation module needs to build.

2. Machine Learning Algorithms for Energy Consumption Prediction in Smart Homes: A Comparative Study

Authors: (Springer book chapter, Studies in Computational Intelligence series) | **Year/Publication:** 2025

Link/DOI: https://link.springer.com/chapter/10.1007/978-3-031-80817-3_5

Problem Solved	Many competing ML algorithms exist for smart-home energy prediction with no consolidated comparison for practitioners choosing a model.
Research Objective	Comparatively evaluate ML algorithms for smart-home electrical energy consumption prediction.
Methodology / Technology	Comparative benchmarking of multiple ML/deep-learning algorithms against smart-home datasets, citing prior recommendation-oriented solar/energy systems.
Dataset / Data Used	Smart home energy consumption datasets (referenced from multiple prior studies).
Main Findings / Results	Different model families suit different data regimes (deep learning for sequence data vs. tree ensembles for tabular); references a solar-energy recommendation-and-prediction precedent.

Limitations / Research Gap	As a comparative survey, doesn't propose a new production-ready model; conclusions depend heavily on dataset choice.
Who Can Use / Benefit	Engineers selecting an algorithm family for a new energy-prediction feature.
Relevance to GreenPlus	Useful as model-selection guidance when GreenPlus's team decides which ML family to use for the Energy Recommendation module.

GreenPlus Module Supported: Module 2: Energy Recommendation

Key Point for Related Work: A practical model-comparison reference to justify GreenPlus's algorithm choice for its Energy Recommendation module.

3. Personalized Residential Energy Usage Recommendation System Based on Load Monitoring and Collaborative Filtering

Authors: Luo, F., Ranzi, G., Kong, W., et al. | **Year/Publication:** 2020, IEEE Transactions on Industrial Informatics, 17(2), 1253-1262

Link/DOI: <https://ieeexplore.ieee.org/document/9078031/>

Problem Solved	Generic energy-saving tips are not personalized to a household's actual usage pattern, reducing their effectiveness.
Research Objective	Build a personalized energy-usage recommendation system combining non-intrusive load monitoring with collaborative filtering.
Methodology / Technology	Load monitoring (disaggregating appliance-level usage) combined with collaborative-filtering-based recommendation.
Dataset / Data Used	Residential smart-meter load-monitoring data.
Main Findings / Results	Personalized, appliance-level recommendations outperform generic tips in relevance and (implied) uptake potential.
Limitations / Research Gap	Requires granular smart-meter/appliance-level data, which not all users/regions have access to.
Who Can Use / Benefit	Utilities and energy-app developers wanting appliance-specific, personalized recommendations rather than generic advice.
Relevance to GreenPlus	This is the closest existing academic precedent to GreenPlus's Energy Recommendation module concept: personalized, data-driven energy advice.

GreenPlus Module Supported: Module 2: Energy Recommendation

Key Point for Related Work: A directly analogous, published system (IEEE) combining load monitoring + collaborative filtering to generate personalized energy recommendations -- GreenPlus's closest technical precedent.

4. Optimizing Daily Household Energy Consumption Prediction Using Ensemble Machine Learning, Feature Selection, and Explainable AI

Authors: (Journal of Intelligent Systems Technology and Informatics) | **Year/Publication:** 2026

Link/DOI: <https://journal.aprika.org/index.php/jistics/article/view/224>

Problem Solved	Household energy models are often 'black boxes,' making it hard for users or developers to trust or act on predictions.
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Research Objective	Combine ensemble ML, feature selection, and explainable AI (XAI) to produce accurate and interpretable household energy predictions.
Methodology / Technology	Ensemble ML models with feature selection and XAI techniques (e.g., SHAP-style explanations) for household-level forecasting.
Dataset / Data Used	Household energy consumption datasets (referenced against Scientific Reports and arXiv baselines).
Main Findings / Results	Explainability layers let users see which factors (time of day, weather, appliance patterns) drive their predicted usage, not just a number.
Limitations / Research Gap	XAI adds computational overhead; explanation quality depends on the underlying ensemble model's structure.
Who Can Use / Benefit	Product teams wanting to show users <i>*why*</i> a recommendation was made, improving trust and adoption.
Relevance to GreenPlus	Explainability is important for GreenPlus's Personalized Eco Advice and Energy Recommendation modules -- users are more likely to act on advice they understand.

GreenPlus Module Supported: Module 2: Energy Recommendation

Key Point for Related Work: Supports adding an explainability layer to GreenPlus's Energy Recommendation output so users understand and trust the suggested actions, not just receive a number.

Module 3: Water Recommendation

1. Advanced Strategies for Monitoring Water Consumption Patterns in Households Based on IoT and Machine Learning

Authors: (MDPI Water, Special Issue: Smart Water Solutions with Big Data) | Year/Publication: 2022, Water, 14(14), 2187

Link/DOI: <https://doi.org/10.3390/w14142187>

Problem Solved	Households lack visibility into which specific water outlets (shower, tap, toilet, etc.) drive their consumption.
Research Objective	Monitor and predict water consumption per outlet using an IoT + ML pipeline, and generate consumer recommendations.
Methodology / Technology	IoT sensor data pipeline with K-means clustering, autonomous disaggregation, and demand-forecasting models.
Dataset / Data Used	Sensor-collected household water-outlet data (flow sensors on taps/showers/appliances).
Main Findings / Results	Per-outlet disaggregation combined with ML enables targeted recommendations, e.g., identifying the highest-waste fixture in a home.
Limitations / Research Gap	Requires physical sensor installation (tedious wiring); limited to households in the pilot study; high raw-data volume.
Who Can Use / Benefit	Smart-home/water-utility developers building fixture-level water-saving tools.
Relevance to GreenPlus	Directly relevant blueprint for a more advanced (sensor-based) version of GreenPlus's Water Recommendation module.

GreenPlus Module Supported: Module 3: Water Recommendation

Key Point for Related Work: Demonstrates outlet-level water monitoring + ML-based recommendations -- a model for a future, sensor-integrated version of GreenPlus's Water module.

2. A Systematic Literature Review on Machine Learning Techniques for Predicting Household Water Consumption

Authors: (Water Resources Management, Springer) | Year/Publication: 2026

Link/DOI: <https://link.springer.com/article/10.1007/s11269-026-04496-4>

Problem Solved	Evidence on which ML models work best for household water forecasting is fragmented across 80+ scattered studies.
Research Objective	Systematically review ML techniques for household water-consumption prediction and provide model-selection guidance.
Methodology / Technology	PRISMA-guided systematic review of 80 primary studies (2009-2024) from ScienceDirect and Web of Science.
Dataset / Data Used	Meta-analysis of 80 studies; no single primary dataset.
Main Findings / Results	LSTM/recurrent models dominate for sequence-based (time-series) forecasting; tree-ensemble models dominate for tabular, property-based prediction.

Limitations / Research Gap	As a review, doesn't test new models directly; conclusions are only as good as the underlying primary studies.
Who Can Use / Benefit	Teams choosing a model architecture for water-consumption forecasting.
Relevance to GreenPlus	Gives GreenPlus evidence-based guidance on whether to use LSTM (if tracking usage over time) or tree ensembles (if using property/household features) for its Water Recommendation module.

GreenPlus Module Supported: Module 3: Water Recommendation

Key Point for Related Work: An 80-study systematic review giving clear, evidence-based model-selection guidance for GreenPlus's Water Recommendation module.

3. Machine Learning for Smart Water Distribution Systems: Exploring Applications, Challenges and Future Perspectives

Authors: (Artificial Intelligence Review, Springer) | **Year/Publication:** 2025

Link/DOI: <https://link.springer.com/article/10.1007/s10462-024-11093-7>

Problem Solved	Existing taxonomies of smart-water-meter analytics are narrow, mostly limited to behavioral analysis or demand forecasting alone.
Research Objective	Propose a broader taxonomy of ML applications across both infrastructural analysis and demand analysis in water systems.
Methodology / Technology	Literature review and taxonomy development from high-resolution smart-meter time-series data patterns.
Dataset / Data Used	Referenced smart-meter datasets across surveyed studies.
Main Findings / Results	High-resolution (hourly) smart-meter data enables both peak-demand prediction and infrastructure-level anomaly detection (e.g., leak detection).
Limitations / Research Gap	Broad taxonomy paper; doesn't validate a specific deployable model.
Who Can Use / Benefit	Utilities and app developers wanting to extend water tools beyond consumption prediction into leak/anomaly detection.
Relevance to GreenPlus	Suggests a future extension for GreenPlus's Water module: leak detection alerts, not just consumption recommendations.

GreenPlus Module Supported: Module 3: Water Recommendation

Key Point for Related Work: Broadens the scope GreenPlus could consider for its Water Recommendation module -- beyond usage tips to leak/anomaly alerts.

4. Predicting and Understanding Residential Water Use with Interpretable Machine Learning

Authors: (Environmental Research Letters, IOP Publishing) | **Year/Publication:** 2024

Link/DOI: <https://iopscience.iop.org/article/10.1088/1748-9326/ad1434>

Problem Solved	Prior approaches to residential water-use drivers were either accurate-but-uninterpretable (non-parametric ML) or interpretable-but-inaccurate (parametric statistics).
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Research Objective	Use interpretable ML to both predict and explain the drivers of residential water use, including behavioral/curtailment factors.
Methodology / Technology	Non-parametric ML models combined with interpretable-ML techniques (e.g., partial dependence / feature attribution).
Dataset / Data Used	Residential water-use records, including a variable for mandated curtailment periods (e.g., drought restrictions).
Main Findings / Results	Interpretable ML captures non-linear feature interactions (better than parametric models) while still explaining *why* usage is high or low -- balancing accuracy and explainability.
Limitations / Research Gap	Study region-specific; curtailment-mandate variable may not generalize outside jurisdictions with such policies.
Who Can Use / Benefit	Water utilities and app developers who need both accurate prediction and explainable output for policy or user communication.
Relevance to GreenPlus	Supports GreenPlus combining accurate water-use prediction with clear, explainable reasons -- important for the 'Personalized Eco Advice' layer on top of the Water module.

GreenPlus Module Supported: Module 3: Water Recommendation

Key Point for Related Work: Shows how to balance prediction accuracy and explainability for water use -- directly useful for GreenPlus's Water Recommendation + Personalized Eco Advice integration.

Module 4: Green Action Prediction

1. Predicting Pro-Environmental Behavioral Intention Using Interpretable Machine Learning

Authors: (Scientific Reports, Nature Portfolio) | **Year/Publication:** 2026

Link/DOI: <https://www.nature.com/articles/s41598-026-52271-7>

Problem Solved	Traditional behavioral-intention models (e.g., Theory of Planned Behavior) are explanatory but weak at individual-level prediction.
Research Objective	Compare 9 ML algorithms for predicting pro-environmental behavioral intention (PEBI) and identify its key drivers.
Methodology / Technology	Benchmarked 9 ML algorithms (Random Forest, etc.) with interpretability analysis of feature importance.
Dataset / Data Used	Survey data on individuals' pro-environmental behavioral intention and psychosocial factors.
Main Findings / Results	Random Forest gave the highest predictive accuracy; connection-to-nature and social-network factors were strong predictors alongside attitudes.
Limitations / Research Gap	Predicts *intention*, not verified real-world action -- the classic intention-behavior gap remains unaddressed.
Who Can Use / Benefit	App/policy designers who want to predict which users are likely (or unlikely) to adopt green actions, to target interventions.
Relevance to GreenPlus	Core technical precedent for GreenPlus's Green Action Prediction module -- shows ML can meaningfully forecast who will act sustainably and why.

GreenPlus Module Supported: Module 4: Green Action Prediction

Key Point for Related Work: A 2026 Nature-family paper showing Random Forest can predict pro-environmental intention with good accuracy from psychosocial + social factors -- the closest direct precedent for GreenPlus's Green Action Prediction module.

2. Examining Digital Nudges to Influence Pro-Environmental Behavior

Authors: Balakrishnan, J., Al-Ahmadi, M. S., Baabdullah, A. M., Dwivedi, Y. K., et al. | **Year/Publication:** 2025, Journal of Computer Information Systems

Link/DOI: <https://www.tandfonline.com/doi/full/10.1080/08874417.2025.2526385>

Problem Solved	It's unclear whether digital nudges actually convert into real pro-environmental behavior, or only intentions.
Research Objective	Empirically test the direct and indirect effects of digital nudging on tourists' pro-environmental intentions and actions.
Methodology / Technology	Two field-based studies with 320 participants; direct/indirect effect modeling.
Dataset / Data Used	320-participant field study dataset (tourism context).

Main Findings / Results	Nudging effectively boosts pro-environmental *intentions*, but the link from intention to *actual behavior* remains weak -- confirming a persistent intention-behavior gap.
Limitations / Research Gap	Tourism-specific context may not generalize to everyday household behavior; intention-behavior gap remains unsolved.
Who Can Use / Benefit	Product teams designing nudge/notification systems who need realistic expectations about nudge effectiveness.
Relevance to GreenPlus	Important caution for GreenPlus: notifications/nudges alone may boost stated intent but not guarantee real action -- suggests pairing predictions with verifiable action tracking or gamification.

GreenPlus Module Supported: Module 4: Green Action Prediction

Key Point for Related Work: Empirically shows digital nudges work for intentions but not reliably for behavior -- a key limitation GreenPlus should design around (e.g., via gamification or verified action logging).

3. Predictive Analysis of the Pro-Environmental Behaviour of College Students Using a Decision-Tree Model

Authors: (PMC / peer-reviewed journal, open access) | Year/Publication: 2022

Link/DOI: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9367762/>

Problem Solved	Need a simple, interpretable model to identify which factors predict pro-environmental behavior in a defined population.
Research Objective	Build and validate a decision-tree model predicting college students' pro-environmental behavior from psychosocial variables.
Methodology / Technology	Decision-tree classification model.
Dataset / Data Used	Survey of college students measuring environmental willingness, activism, subjective norms, sense of place, social identity, etc.
Main Findings / Results	Willingness to act, environmental activism, and subjective norms were the top predictors, in that order; decision tree was a valid predictive tool.
Limitations / Research Gap	Single population (college students); decision trees can overfit without pruning/cross-validation reporting.
Who Can Use / Benefit	Educators/app designers targeting a specific demographic (e.g., students) with tailored green-action prompts.
Relevance to GreenPlus	A simple, interpretable model architecture (decision tree) GreenPlus could use for an early, lightweight version of Green Action Prediction before scaling to more complex models.

GreenPlus Module Supported: Module 4: Green Action Prediction

Key Point for Related Work: Shows a simple decision-tree model can meaningfully predict green behavior from a handful of psychosocial features -- a good lightweight starting architecture for GreenPlus.

4. LLM-Generated Personalized Nudges for Improving Pro-Environmental Behavior: Field Evidence from Resource Conservation

Authors: (arXiv preprint) | Year/Publication: 2026

Link/DOI: <https://arxiv.org/html/2604.03881v2>

Problem Solved	Generic, one-size-fits-all nudges underperform because they ignore individual differences in baseline behavior and psychology.
Research Objective	Test whether LLM-personalized nudges (via chatbot) outperform conventional nudges at reducing electricity and hot-water use.
Methodology / Technology	Field experiment; XGBoost models used to analyze which baseline characteristics predict intervention-period consumption; chatbot-delivered personalized nudges.
Dataset / Data Used	Field trial data: daily electricity and hot-water use, chatbot interaction logs, baseline psychological measures.
Main Findings / Results	Personalization (via LLM-generated nudges) improved resource-conservation outcomes over generic nudges; ML helped identify which user segments respond best.
Limitations / Research Gap	Requires ongoing chatbot infrastructure and LLM costs; field trial scale is likely modest (typical for this study type).
Who Can Use / Benefit	Product teams designing AI-chatbot-based, personalized sustainability coaching (very close to GreenPlus's Personalized Eco Advice concept).
Relevance to GreenPlus	Near-direct precedent for combining GreenPlus's Green Action Prediction with Personalized Eco Advice via a conversational/LLM interface.

GreenPlus Module Supported: Module 4: Green Action Prediction

Key Point for Related Work: Cutting-edge (2026) evidence that LLM-personalized nudges outperform generic ones for real resource conservation -- a strong technical justification for GreenPlus's personalized, chat-style advice direction.

Module 5: Personalized Eco Advice

1. Advancing Sustainability via Recommender Systems: A Survey

Authors: Zhou, X., Zhang, L., Zhang, H., Zhang, Y., Zhang, X., Zhang, J., Shen, Z. | **Year/Publication:** 2025, arXiv preprint (Nanyang Technological University)

Link/DOI: <https://arxiv.org/pdf/2411.07658>

Problem Solved	Conventional recommender systems optimize for engagement/revenue, inadvertently reinforcing overconsumption and unsustainable choices.
Research Objective	Systematically survey how recommender systems can be redesigned to advance sustainability goals across domains.
Methodology / Technology	Systematic survey and taxonomy across transportation, food, buildings, and auxiliary sectors.
Dataset / Data Used	None (survey of the sustainable-recommender-systems literature).
Main Findings / Results	Sustainable RS design requires embedding environmental/social objectives directly into the recommendation objective function, not just as a filter after the fact.
Limitations / Research Gap	Survey-level; doesn't propose a single best architecture; domain-specific trade-offs remain open research questions.
Who Can Use / Benefit	Anyone designing a recommender system meant to shift consumption behavior sustainably -- directly applicable to GreenPlus.
Relevance to GreenPlus	This is the master reference for GreenPlus's entire Personalized Eco Advice module -- explicitly frames the field GreenPlus operates in.

GreenPlus Module Supported: Module 5: Personalized Eco Advice

Key Point for Related Work: The definitive, up-to-date survey establishing 'sustainable recommender systems' as a field -- the anchor citation for GreenPlus's Personalized Eco Advice module.

2. Preference Learning for Eco-Friendly Hotels Recommendation: A Multi-Criteria Collaborative Filtering Approach

Authors: Nilashi, M., Ahani, A., Esfahani, M. D., et al. | **Year/Publication:** 2019, Journal of Cleaner Production, 215, 767-783

Link/DOI: <https://doi.org/10.1016/j.jclepro.2019.01.012>

Problem Solved	Standard hotel-recommendation systems don't factor in environmental/eco-friendliness criteria alongside standard preferences.
Research Objective	Build a multi-criteria collaborative filtering system that recommends eco-friendly hotels aligned with user preferences.
Methodology / Technology	Multi-criteria collaborative filtering combining preference learning with sustainability criteria.
Dataset / Data Used	Hotel review/preference dataset with eco-friendliness attributes.
Main Findings / Results	Multi-criteria filtering can successfully surface eco-friendly options without sacrificing personalization/relevance to the user.

Limitations / Research Gap	Domain-specific to hospitality; eco-criteria weighting choices are somewhat subjective.
Who Can Use / Benefit	E-commerce/travel platforms wanting to blend personalization with sustainability filtering.
Relevance to GreenPlus	A proven pattern (multi-criteria filtering: personal fit + eco-score) directly transferable to GreenPlus's Personalized Eco Advice for products/services beyond hotels.

GreenPlus Module Supported: Module 5: Personalized Eco Advice

Key Point for Related Work: An early (2019), well-cited example of blending personalization with an explicit eco-criterion -- a reusable pattern for GreenPlus's advice engine.

3. Recommender Systems and Sustainability: A Dual Perspective

Authors: (ScienceDirect journal article) | **Year/Publication:** 2026

Link/DOI: <https://www.sciencedirect.com/science/article/pii/S1574013726000213>

Problem Solved	Most sustainability-oriented recommender research targets a single SDG in isolation, missing opportunities for systems that advance multiple SDGs at once.
Research Objective	Analyze how recommender systems can be designed to simultaneously advance multiple Sustainable Development Goals.
Methodology / Technology	Conceptual/analytical dual-perspective framework mapping RS design choices to multiple SDGs.
Dataset / Data Used	None (conceptual framework paper).
Main Findings / Results	RS applications addressing SDG 12 (responsible consumption) can also intersect environmental and economic pillars simultaneously if designed deliberately.
Limitations / Research Gap	Conceptual, not empirically validated with a deployed system.
Who Can Use / Benefit	Strategic product teams wanting their platform to be positioned against multiple SDGs (e.g., 6, 7, 12, 13) rather than just one.
Relevance to GreenPlus	Supports GreenPlus's multi-module design (carbon + energy + water + behavior) as a deliberate, SDG-spanning strategy rather than a single-issue tool.

GreenPlus Module Supported: Module 5: Personalized Eco Advice

Key Point for Related Work: Provides the strategic framing for why GreenPlus's multi-module approach (carbon, energy, water, behavior together) is a stronger design than a single-issue tool.

4. Understanding the Impact of Sustainability-Oriented Recommender Systems on Consumers' Choices

Authors: (ScienceDirect journal article) | **Year/Publication:** 2025

Link/DOI: <https://www.sciencedirect.com/science/article/abs/pii/S1567422325000651>

Problem Solved	It's unclear whether sustainability-oriented recommendations actually change consumer choices or are simply ignored like other nudges.
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Research Objective	Examine the real impact of sustainability-oriented recommender systems on consumer decision-making.
Methodology / Technology	Empirical study of consumer response to sustainability-labeled recommendations (platform-based).
Dataset / Data Used	Consumer choice/interaction data from a digital retail platform.
Main Findings / Results	Sustainability-oriented recommendations can shift consumer choices while preserving platform engagement/revenue -- personalization and sustainability aren't necessarily in tension.
Limitations / Research Gap	Platform- and market-specific; may not generalize to non-commercial/behavior-only apps like GreenPlus.
Who Can Use / Benefit	Platforms wanting evidence that adding a sustainability signal to recommendations won't hurt user engagement.
Relevance to GreenPlus	Reassures GreenPlus's design team that layering eco-advice on top of recommendations doesn't necessarily reduce user engagement or satisfaction.

GreenPlus Module Supported: Module 5: Personalized Eco Advice

Key Point for Related Work: Empirical evidence that sustainability-oriented recommendations can shift real consumer choices without harming engagement -- de-risks GreenPlus's core value proposition.

Module 6 (Later): Methane Model

1. Approaches for Predicting Dairy Cattle Methane Emissions: From Traditional Methods to Machine Learning

Authors: (Journal of Animal Science, Oxford Academic) | Year/Publication: 2024

Link/DOI: <https://academic.oup.com/jas/article/doi/10.1093/jas/skae219/7731169>

Problem Solved	Direct measurement of dairy cattle methane (CH ₄) emissions is complex and expensive, limiting large-scale monitoring.
Research Objective	Review and compare mechanistic, empirical, and ML-based approaches for predicting dairy cattle methane emissions from proxy data.
Methodology / Technology	Systematic review of prediction-modeling literature (mechanistic vs. empirical vs. ML models).
Dataset / Data Used	Aggregated across many primary studies using proxy data (diet, milk-fatty-acid profiles via MIRS, etc.).
Main Findings / Results	Agriculture is responsible for ~40% of anthropogenic methane, with ~80% from enteric fermentation in ruminants; ML models can predict CH ₄ from cheaper proxy data (e.g., milk composition) instead of direct measurement.
Limitations / Research Gap	ML models still need robust, standardized validation across farms/regions before commercial-scale trust; international data consortia recommended.
Who Can Use / Benefit	Agri-tech developers and researchers building farm-level methane monitoring/mitigation tools.
Relevance to GreenPlus	Foundational technical review for a future GreenPlus Methane Model -- shows proxy-data-based ML prediction is a validated, active research direction.

GreenPlus Module Supported: Module 6 (Later): Methane Model

Key Point for Related Work: A comprehensive review confirming ML-based methane prediction from cheap proxy data is viable and increasingly research-validated -- the technical foundation for GreenPlus's planned Methane Model.

2. Predicting Methane Emissions in Smallholder Dairy Systems: A Clustering and Ensemble Learning Approach

Authors: Choo, W., Raghavan, S. | Year/Publication: 2025, Frontiers in Sustainable Food Systems, 9:1668517

Link/DOI: <https://doi.org/10.3389/fsufs.2025.1668517>

Problem Solved	Most methane-prediction research targets industrial-scale dairy farms; smallholder farms (common in developing countries) are underserved.
Research Objective	Develop a clustering + ensemble-learning approach to forecast methane emissions in smallholder dairy systems.
Methodology / Technology	Clustering combined with a hybrid stacked ensemble ML model; evaluated via time-based train-test splits and leave-one-out cross-validation.
Dataset / Data Used	Smallholder dairy farm data from Indonesia.

Main Findings / Results	The hybrid stacked ensemble outperformed individual models, particularly under leave-one-out cross-validation, showing good generalizability for smallholder contexts.
Limitations / Research Gap	Region-specific (Indonesia); smallholder data is inherently noisier/sparser than industrial-farm data.
Who Can Use / Benefit	NGOs, agri-tech developers, and governments supporting smallholder farmers in emissions monitoring and mitigation.
Relevance to GreenPlus	Especially relevant if GreenPlus targets users in developing regions/smallholder contexts for its future Methane Model.

GreenPlus Module Supported: Module 6 (Later): Methane Model

Key Point for Related Work: Shows ensemble ML can forecast methane emissions even in data-sparse smallholder farm contexts -- important if GreenPlus's Methane Model targets non-industrial users.

3. Enteric Methane Emission in Livestock Sector: Bibliometric Research from 1986 to 2024 with Text Mining and Topic Analysis by Machine Learning Algorithms

Authors: (PMC / open-access peer-reviewed journal) | **Year/Publication:** 2024

Link/DOI: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11545165/>

Problem Solved	No consolidated view existed of how methane-emissions-from-livestock research has evolved over nearly 40 years.
Research Objective	Use text mining and ML-based topic analysis to map the evolution of livestock methane research from 1986-2024.
Methodology / Technology	Bibliometric analysis with ML-driven text mining and topic modeling on 1,294 studies.
Dataset / Data Used	1,294 published studies (1986-2024) on livestock methane emissions.
Main Findings / Results	Research volume increased sharply after 2005, concentrated in Europe/North America; dominant topics are dietary changes and farm-practice improvements to reduce emissions.
Limitations / Research Gap	Bibliometric/meta-level analysis; doesn't itself propose a new prediction model.
Who Can Use / Benefit	Researchers and product teams needing a map of the state-of-the-art and research gaps before building a methane feature.
Relevance to GreenPlus	Useful for GreenPlus's 'research gap' framing: shows where prior work concentrates (diet/farm-practice) and where a consumer-facing predictive tool could add new value.

GreenPlus Module Supported: Module 6 (Later): Methane Model

Key Point for Related Work: A large-scale bibliometric map of 40 years of methane research -- useful to justify the research gap a future consumer-facing GreenPlus Methane Model would fill.

4. An IoT-Enabled Framework for Real-Time Monitoring and Prediction of Methane Emissions in Sustainable Ruminant Farming

Authors: Ezeofor, C. J., Onwuchekwa, N. P. | **Year/Publication:** 2026, International Journal of Inventive Engineering and Sciences (IJIES)

Link/DOI: <https://www.ijies.org/portfolio-item/B105505020226/>

Problem Solved	Conventional methane measurement on farms is manual, slow, and lacks predictive/real-time capability.
Research Objective	Build an AI + IoT framework for real-time methane monitoring and prediction in ruminant farming.
Methodology / Technology	IoT sensing (methane sensor, ATmega328P microcontroller, ESP8266 Wi-Fi, cloud dashboard) combined with Random Forest, XGBoost, and ANN models.
Dataset / Data Used	Feed composition, animal weight, and environmental variables from ruminant farms in South-South Nigeria.
Main Findings / Results	Random Forest performed best (93% accuracy, MAE=1.52, RMSE=2.21); system enables real-time monitoring and predictive alerts, not just static measurement.
Limitations / Research Gap	Simulation-based validation (Proteus); real-farm, large-scale deployment still needed.
Who Can Use / Benefit	Farmers, agri-tech companies, and sustainability-tool developers wanting real-time methane dashboards.
Relevance to GreenPlus	A concrete IoT+ML reference architecture GreenPlus could adapt if it builds real-time sensor integration into its future Methane Model.

GreenPlus Module Supported: Module 6 (Later): Methane Model

Key Point for Related Work: A working IoT+ML reference architecture for real-time methane prediction -- a concrete technical template for GreenPlus's future Methane Model if sensor integration is planned.

Authoritative Resources (Not Peer-Reviewed Research Papers)

These are official reports and treaty bodies -- essential for GreenPlus's credibility and for the Rio Trio education module, but they should NOT be cited as 'research papers' in a bibliography. Cite them as reports / institutional sources instead.

1. Climate Change 2023: Synthesis Report

Organization: IPCC | Year: 2023

Link: <https://www.ipcc.ch/report/ar6/syr/>

The definitive global scientific consensus on climate change causes, impacts, and mitigation pathways. Use for factual grounding, not as a 'research paper' citation -- it's a policy/science synthesis report.

2. United Nations Framework Convention on Climate Change (UNFCCC)

Organization: United Nations | Year: Ongoing treaty body

Link: <https://unfccc.int/>

The primary international treaty framework for climate action. Useful for GreenPlus's Rio Trio module as the policy backbone for climate content.

3. Convention on Biological Diversity (CBD)

Organization: United Nations | Year: Ongoing treaty body

Link: <https://www.cbd.int/convention/>

One of the three 'Rio Conventions.' Provides authoritative biodiversity content for the Rio Trio education module.

4. United Nations Convention to Combat Desertification (UNCCD)

Organization: United Nations | Year: Ongoing treaty body

Link: <https://www.unccd.int/>

The third 'Rio Convention,' completing the Rio Trio (Climate + Biodiversity + Desertification) educational framing.

5. Global Methane Assessment: Benefits and Costs of Mitigating Methane Emissions

Organization: UNEP & Climate and Clean Air Coalition (CCAC) | Year: 2021

Link: <https://www.unep.org/resources/report/global-methane-assessment-benefits-and-costs-mitigating-methane-emissions>

Authoritative global data source for methane mitigation targets and costs -- use for GreenPlus's Methane Model context, not as a peer-reviewed citation.

6. Global Methane Tracker

Organization: International Energy Agency (IEA) | Year: 2025 edition

Link: <https://www.iea.org/reports/global-methane-tracker-2025>

Authoritative, regularly updated methane-emissions data tracker; a good real-time data-reference source for a future GreenPlus Methane Model.
